ThinkRook
By Thinknyx Technologies LLP

MCP Bootcamp: Model Context Protocol with n8n & Claude

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MCP N8N course

Founder and Senior Solutions Architect

Thinknyx®

ThinkRook
By Thinknyx Technologies LLP

Yogesh Raheja

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Why MCP

Why now

Journey we're going to take together

What you need to know before starting

WHY MCP, WHY NOW

Check a calendar

Read a document

Query a database

Send a message

WHY MCP, WHY NOW

- Rebuilt from scratch each time
- Broken again the moment an API changed

WHY MCP, WHY NOW

AI

Model Context Protocol

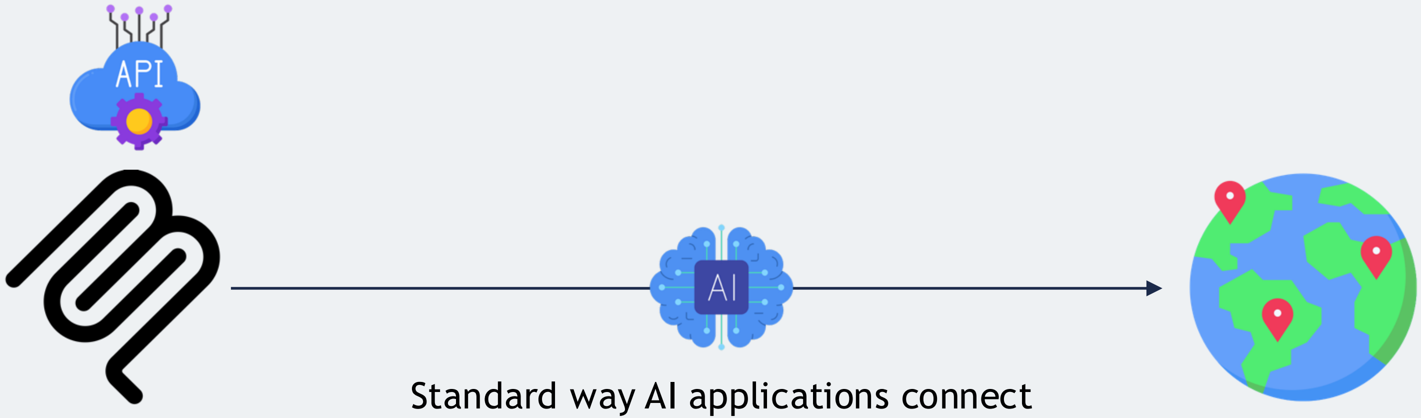
"USB-C for AI"

Open standard that lets any MCP-compatible AI application talk to any MCP-compatible tool

WHY MCP, WHY NOW

MCP has already gone through several major updates in a short time

WHY MCP, WHY NOW



Understanding APIs a decade ago

Standard way AI applications connect to the outside world

THE LEARNING ARC

MCP FUNDAMENTALS

n8n PLATFORM REFRESHER

MCP WITH n8n

n8n + CLAUDE VIA MCP

THE LEARNING ARC

MCP FUNDAMENTALS

- No code, No platform
- What MCP actually is and why it exists
- Host / Client / Server architecture
- Three core primitives
- How communication happens under the hood
- Evolves through versions and deprecations

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- Learn core workflow building concepts
- Practice quickly with MCP demos

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MCP WITH n8n

- Build MCP servers inside n8n
- Expose Google Calendar as tools
- Connect AWS S3 as tools
- Integrate Google Drive with agents
- Connect publicly available MCP servers
- Use Context7 and DeepWiki freely

n8n + CLAUDE VIA MCP

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n8n + CLAUDE VIA MCP

- Connect Claude directly to n8n
- Connect directly over MCP
- Reference your workflows
- Create and publish workflows
- Troubleshoot your workflows
- Claude can work with n8n

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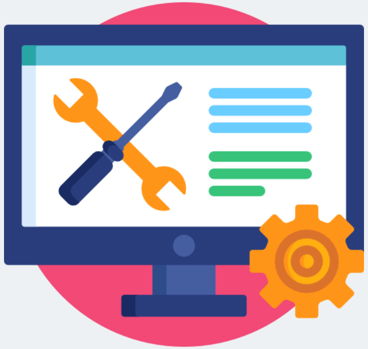
WHAT YOU DO NOT Need

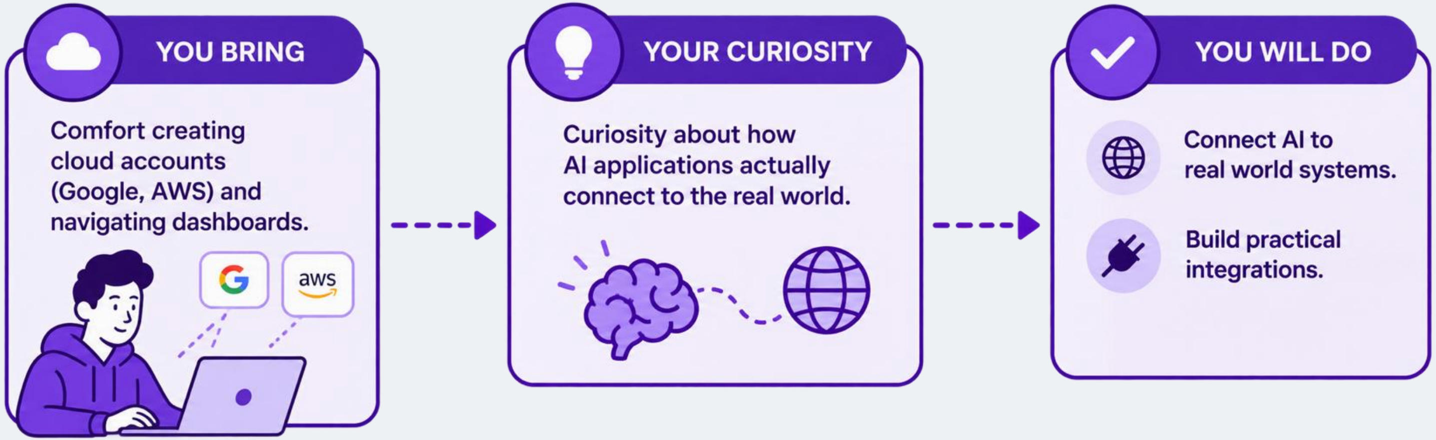
No Prior programming experience

A programming background



Any paid tools to get started

Here are few things What will help



Important Note

n8n Fundamentals

Sections 2 through 4 are a fast refresher

Important Note

n8n Fundamentals

Sections 2 through 4 are a fast refresher

n8n

- Credentials
- Triggers
- Logic
- Error handling

TOOL SETUP

AWS Account

Gmail Id

Claude Desktop

Context7 API Key

MCP is what turns an AI assistant from something that only talks into something that actually acts on your **calendar**, your **files**, your **infrastructure**, your **documentation**

MCP Bootcamp:
Model Context Protocol with n8n & Claude

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MCP n8n Course

COURSE INTRODUCTION

Introduction to MCP n8n

Yogesh Raheja · Thinknyx Technologies LLP

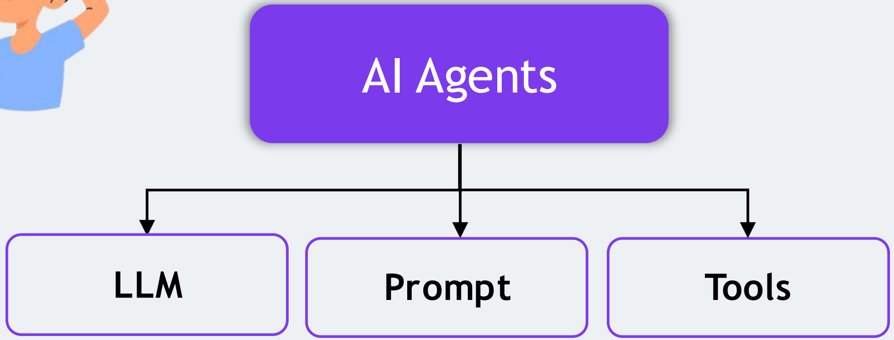
MCP n8n

What is MCP and Why Do We Need It?

Section 1 › Foundations of AI Agents

What is MCP and Why Do We Need It?

What happens when your AI assistant needs to actually do something



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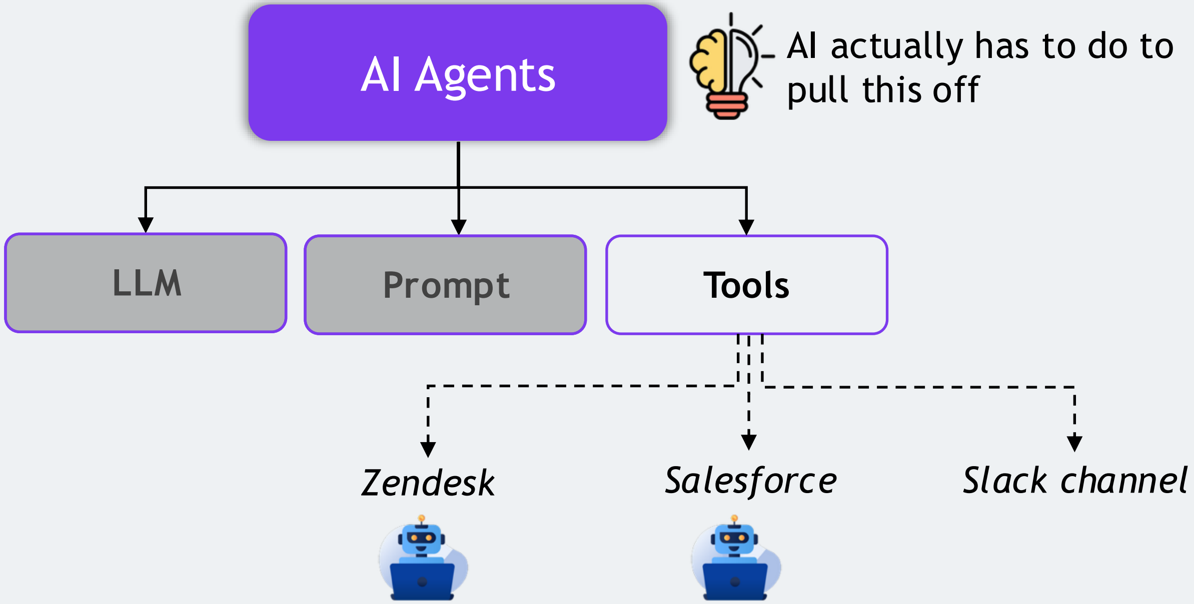

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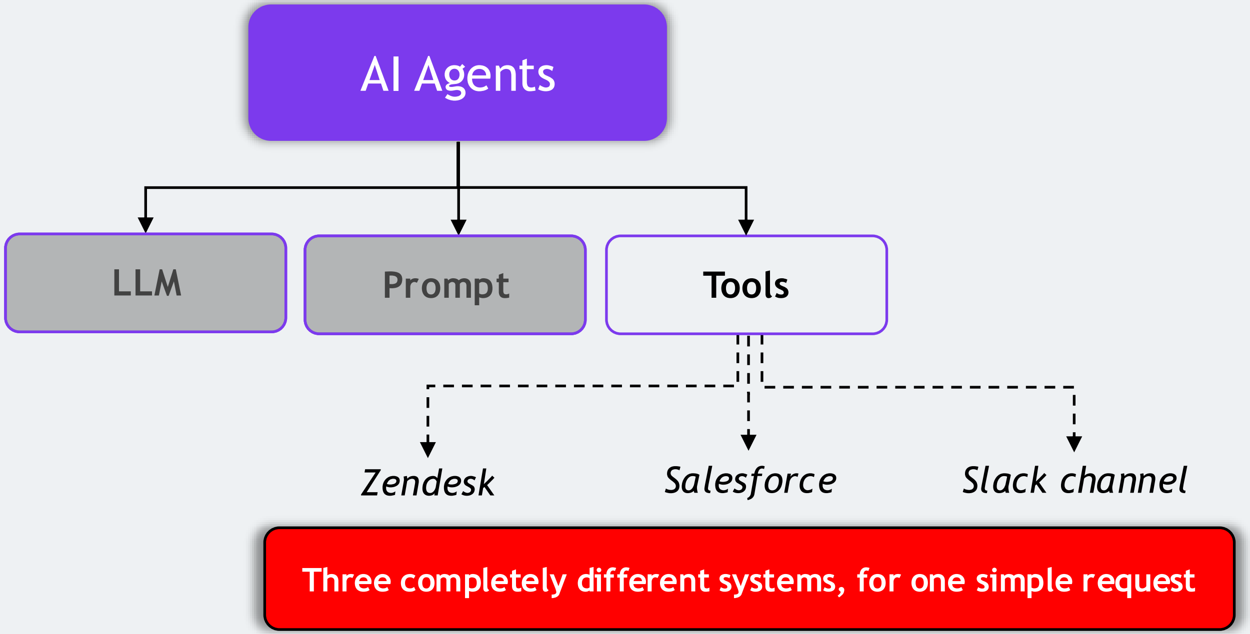


What is MCP and Why Do We Need It?

AI Agents

LLM | Prompt | Tools

Build a separate integration

- For every tool
- For every application
- For every AI assistant

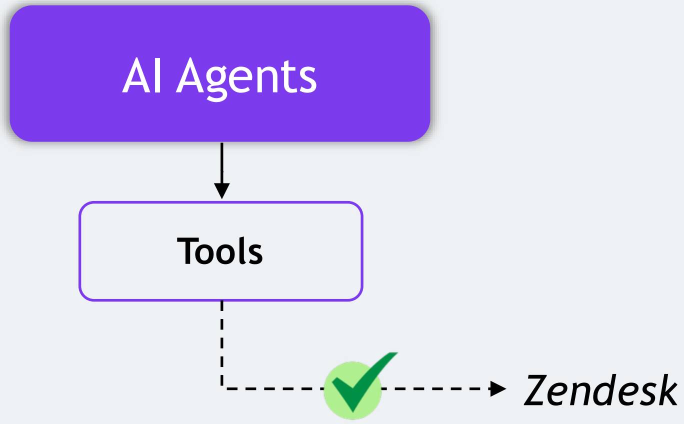
How does one AI application talk to all of these different systems?

What is MCP and Why Do We Need It?

MCP

Model Context Protocol

Let’s understand the first Problem, One Tool at a Time



- Write the integration
- Handle the authentication
- Map the request and response formats

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Let’s understand the first Problem, One Tool at a Time

AI Agents

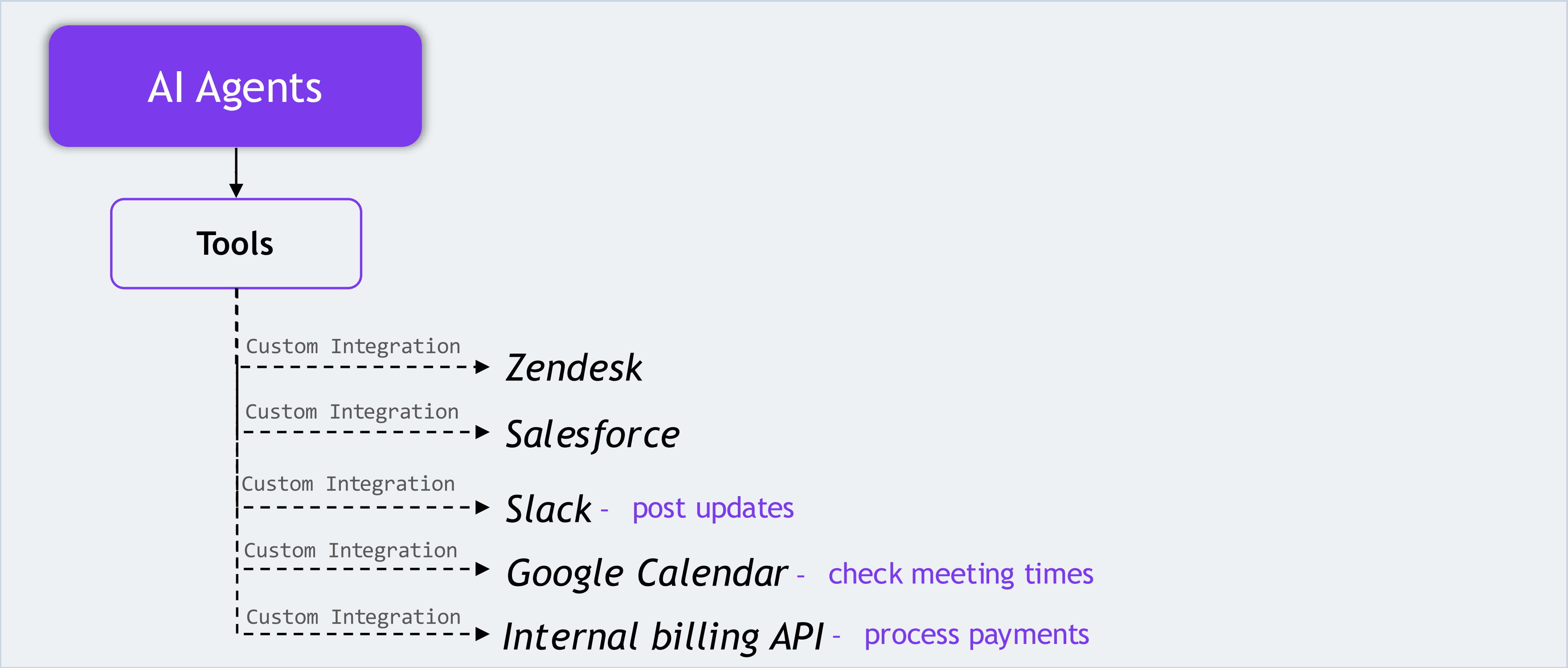
Tools

Zendesk

Salesforce

- Agent can pull customer account details
- Different API, different auth method
- Different response format

Let’s understand the first Problem, One Tool at a Time



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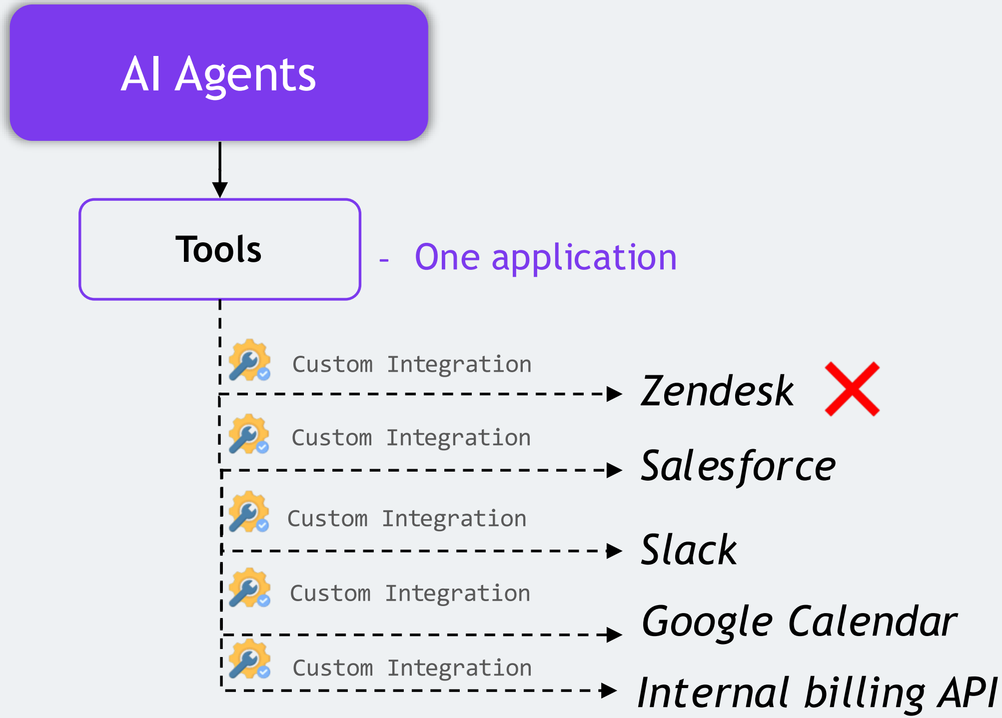
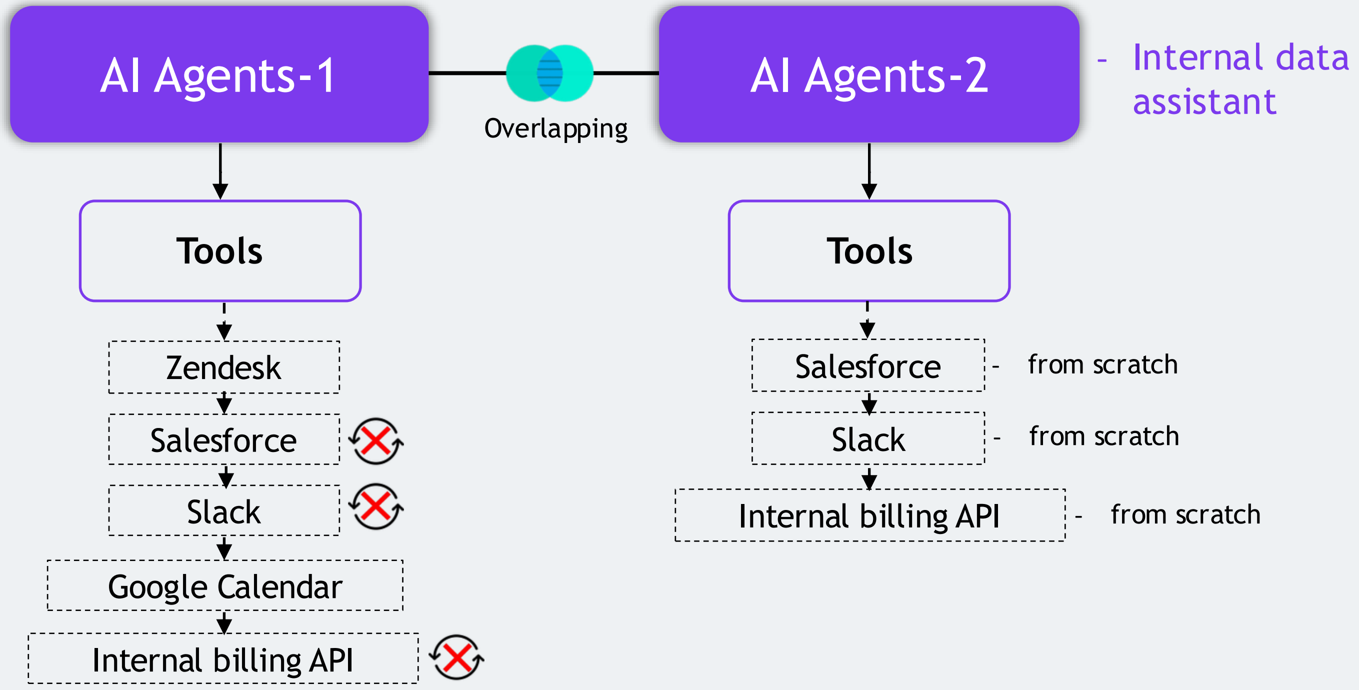


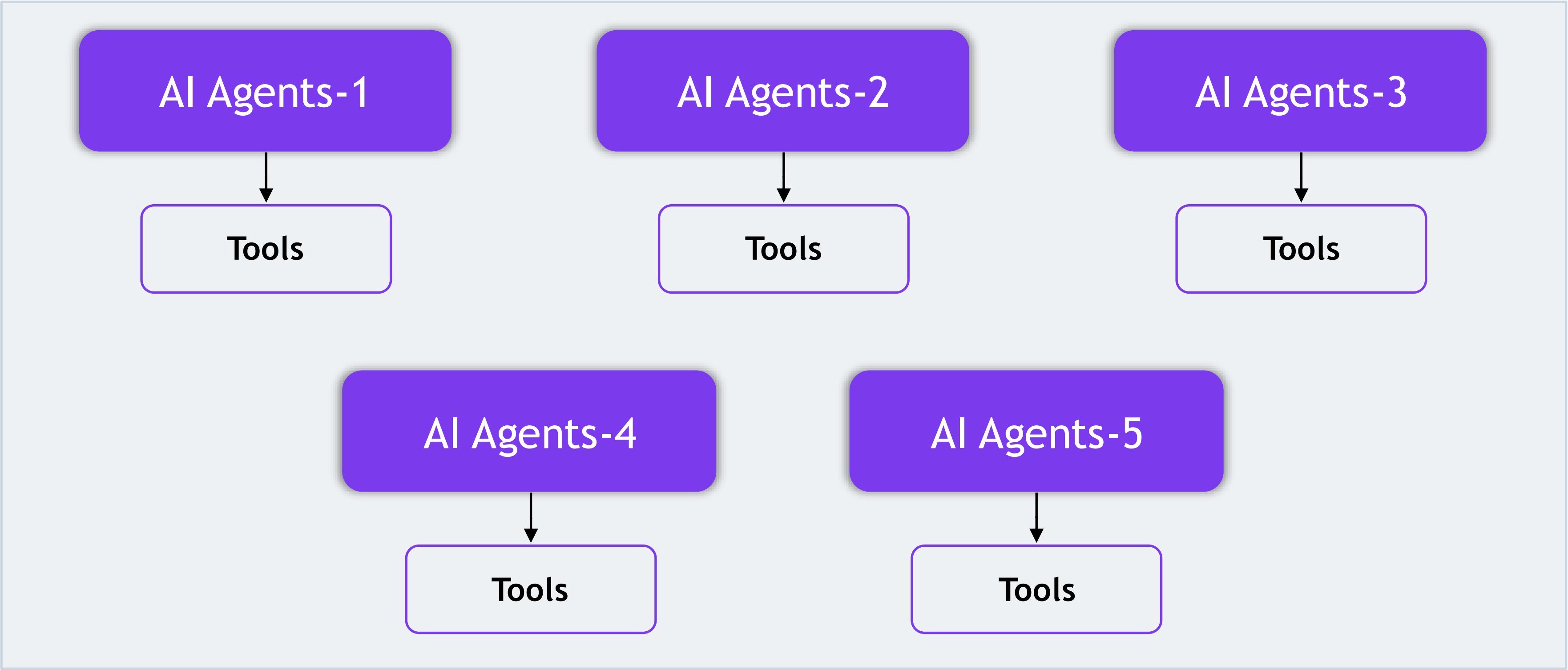
Separate custom-built integration

- Different API
- Different authentication
- Different request and response format

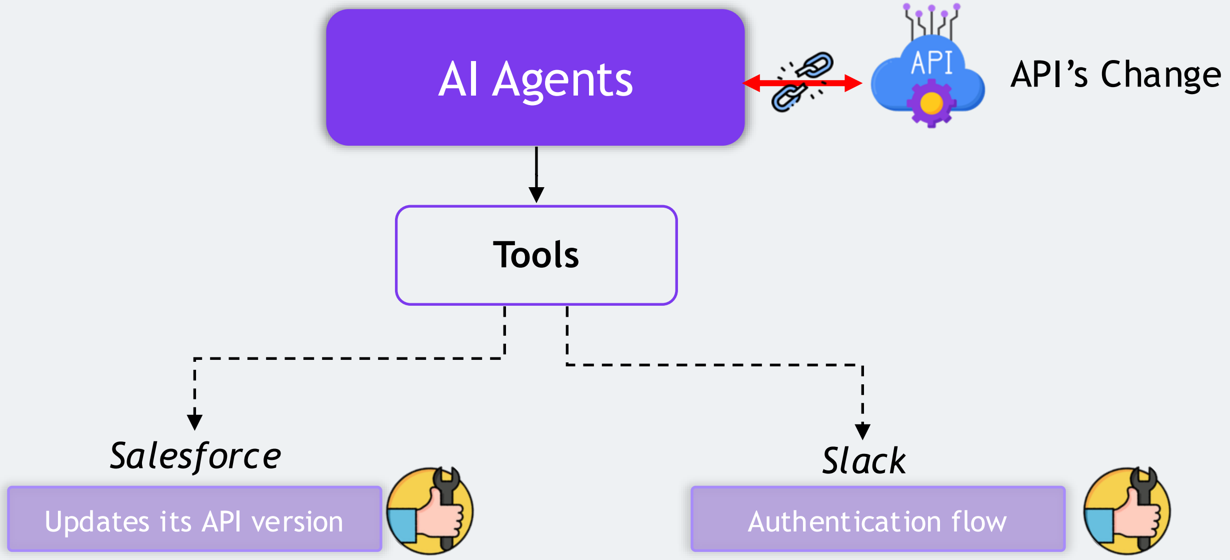
Now Multiply the Problem



Now Multiply the Problem



Then the APIs Change



Then the APIs Change

Three compounding problems

- Every tool needs custom work
- Every new agent repeats that work
- Every API change forces you to redo it

What Exactly Is MCP?

Model Context Protocol

MCP is an open standard that allows AI applications to connect with external systems data sources, tools, and workflows.

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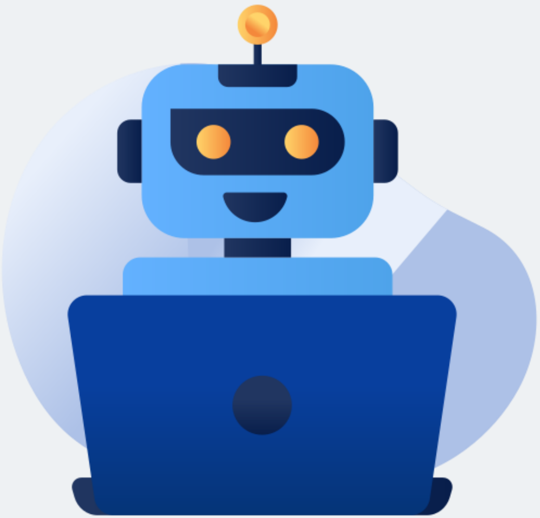
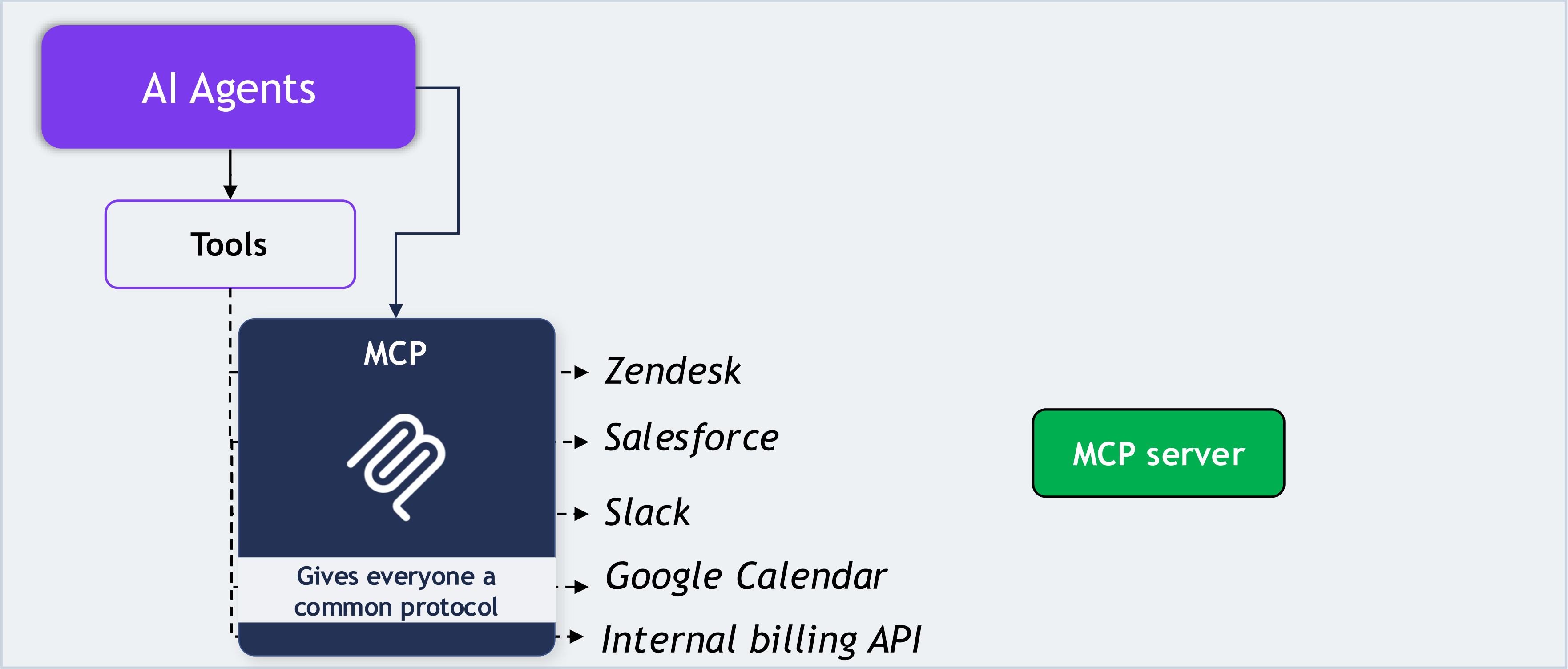
What Exactly Is MCP?



- AI can generate responses using its existing knowledge and context
- External actions require access to outside systems and data
- MCP provides a standardized way to connect and communicate with them

MCP Provides a Standard



The USB-C Analogy

USB-C

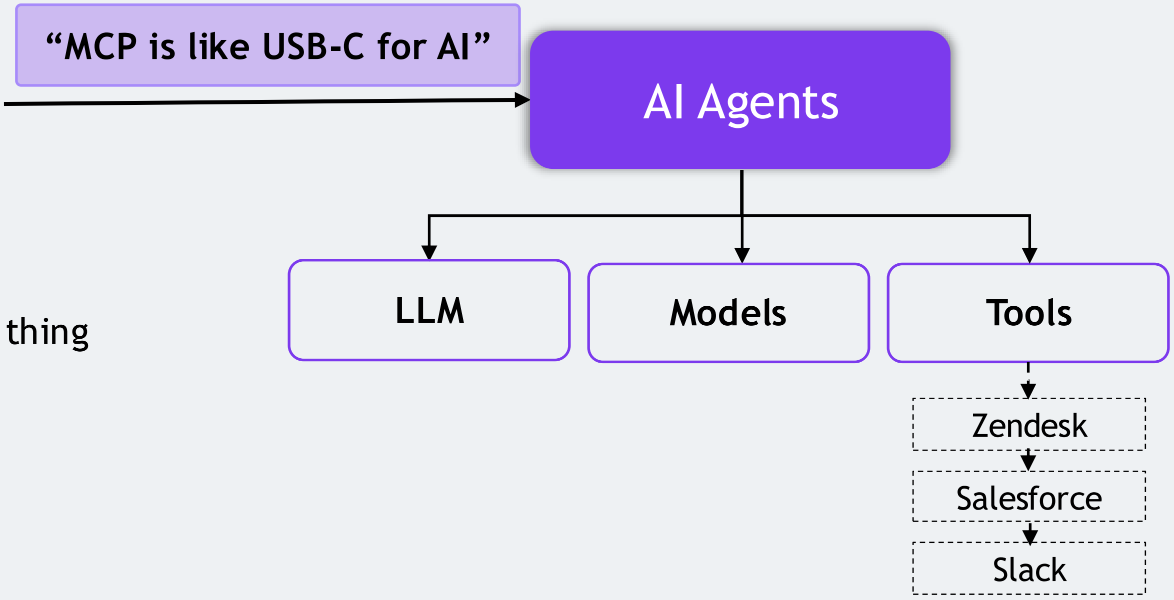
One connector works across
phones, laptops, tablets, cameras,
almost everything

Own connector

Own cable

Own charger

The USB-C Analogy



MCP does the same thing

What Can MCP Connect To?

MCP isn't limited to one type of system

AI application connected through MCP can work with

Data sources

Files, databases, documents, knowledge bases

Tools

Search engines, calculators, APIs, code execution tools, business applications

Workflows & Apps

Calendar, Salesforce, Slack, Notion, Figma

Why Do We Need MCP? — Three Reasons

Standardization

Reusability

Ecosystem

Why Do We Need MCP? — Three Reasons

Standardization

Reusability

Ecosystem

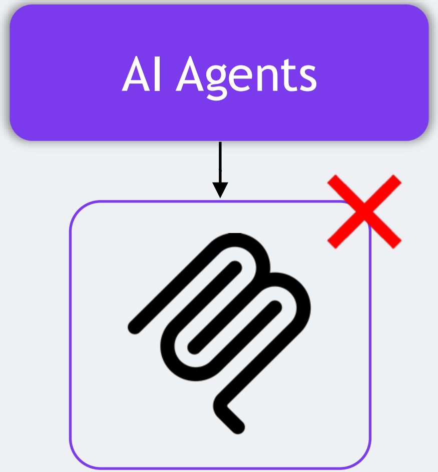
Why Do We Need MCP? — Three Reasons

Standardization

Reusability

Ecosystem

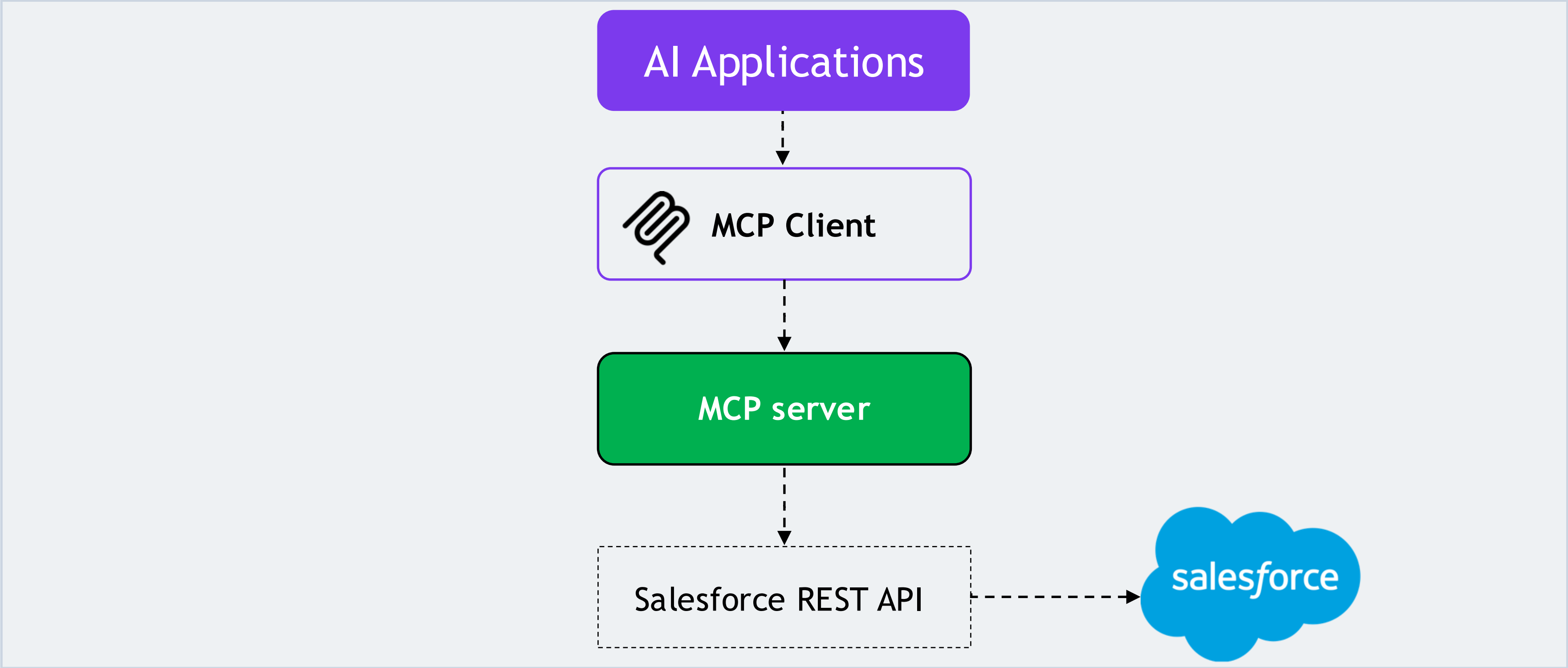
MCP Does Not Replace APIs



MCP replaces REST APIs?

MCP and APIs solve different problems, and an MCP server very often uses an existing API internally

MCP Does Not Replace APIs



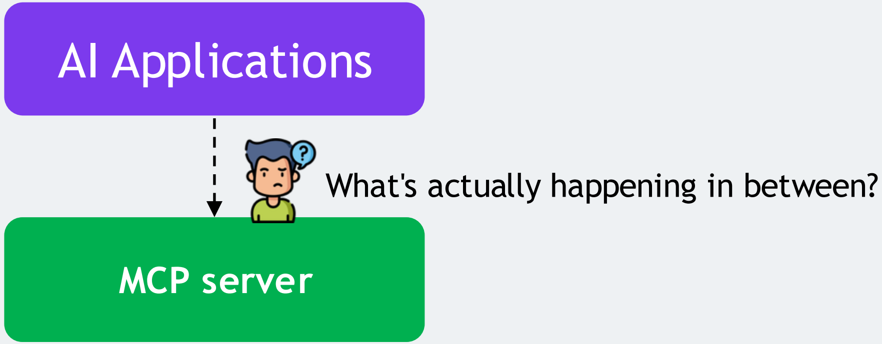
MCP Architecture: Host, Client & Server

Section 2 › MCP Architecture

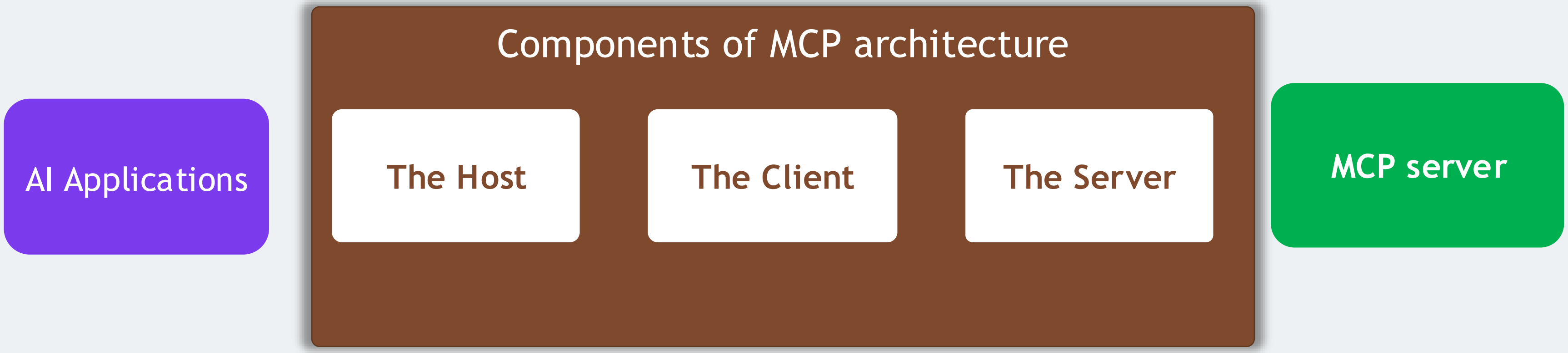
MCP Architecture

Who actually talks to whom?

MCP Architecture



MCP Architecture

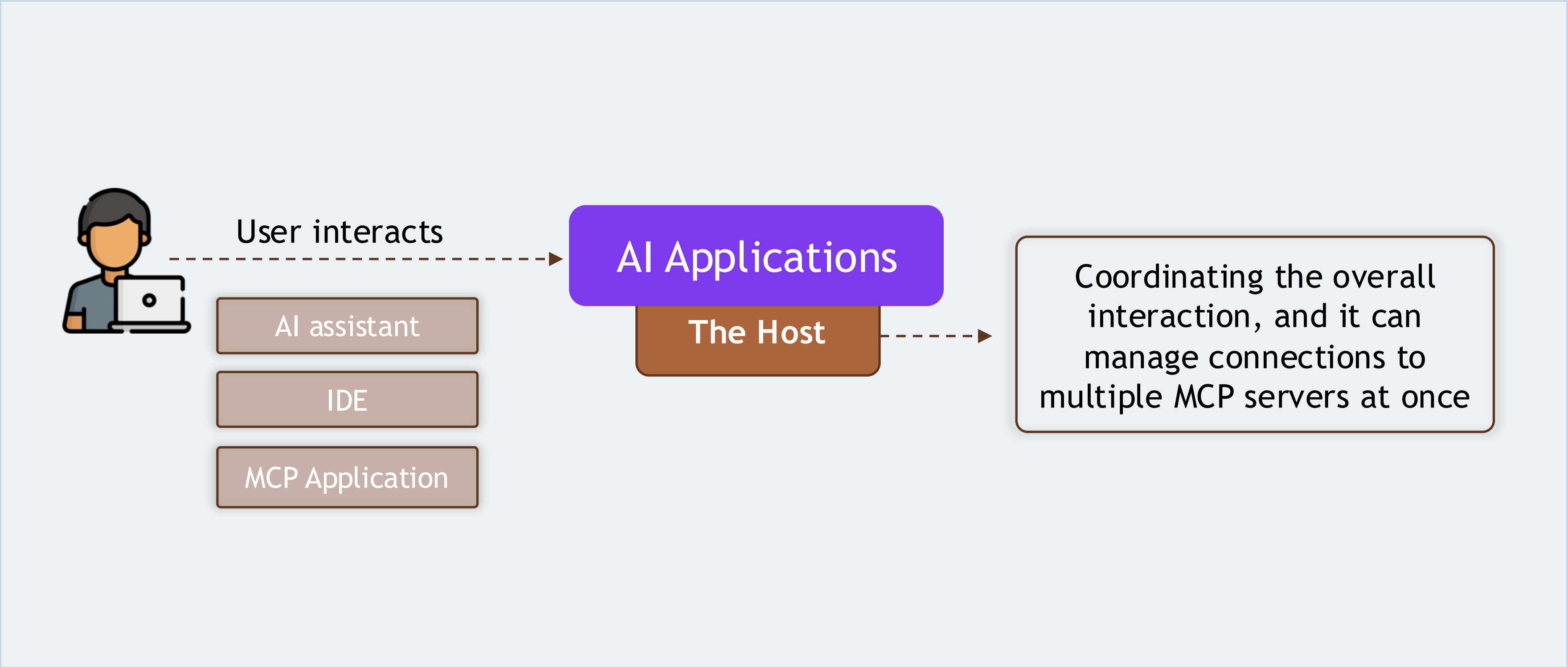


MCP Architecture

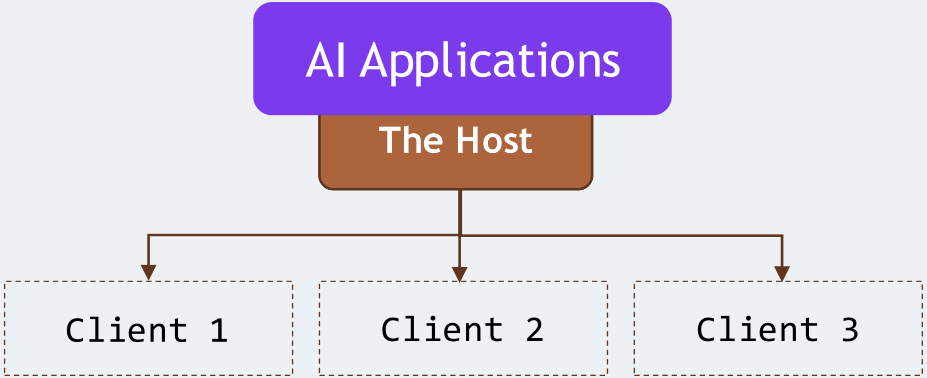
Does the AI model talk to the server directly?

MCP defines a specific architecture
with three distinct roles

MCP Host

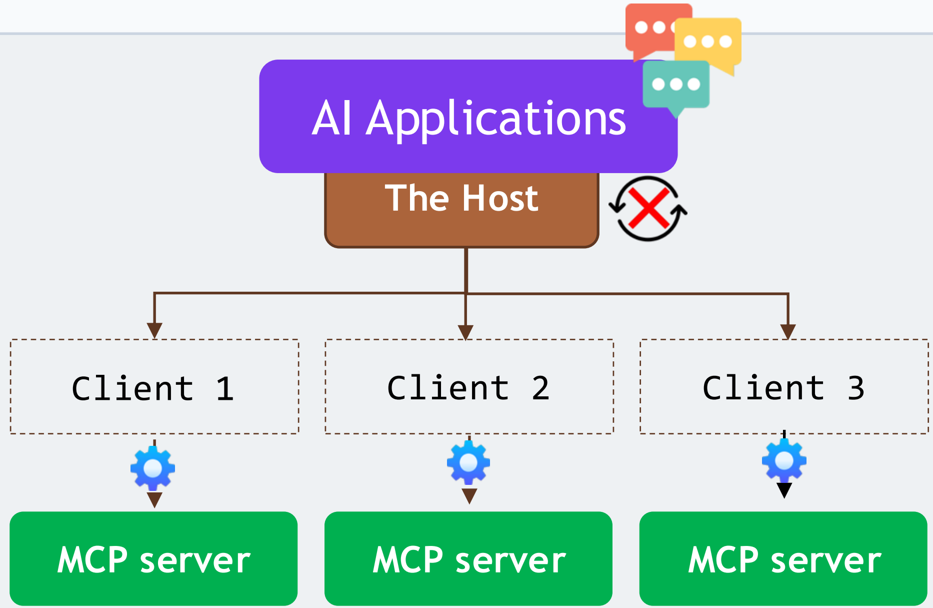

User interacts
AI Applications
The Host
AI assistant
IDE
MCP Application
Coordinating the overall interaction, and it can manage connections to multiple MCP servers at once

MCP Client



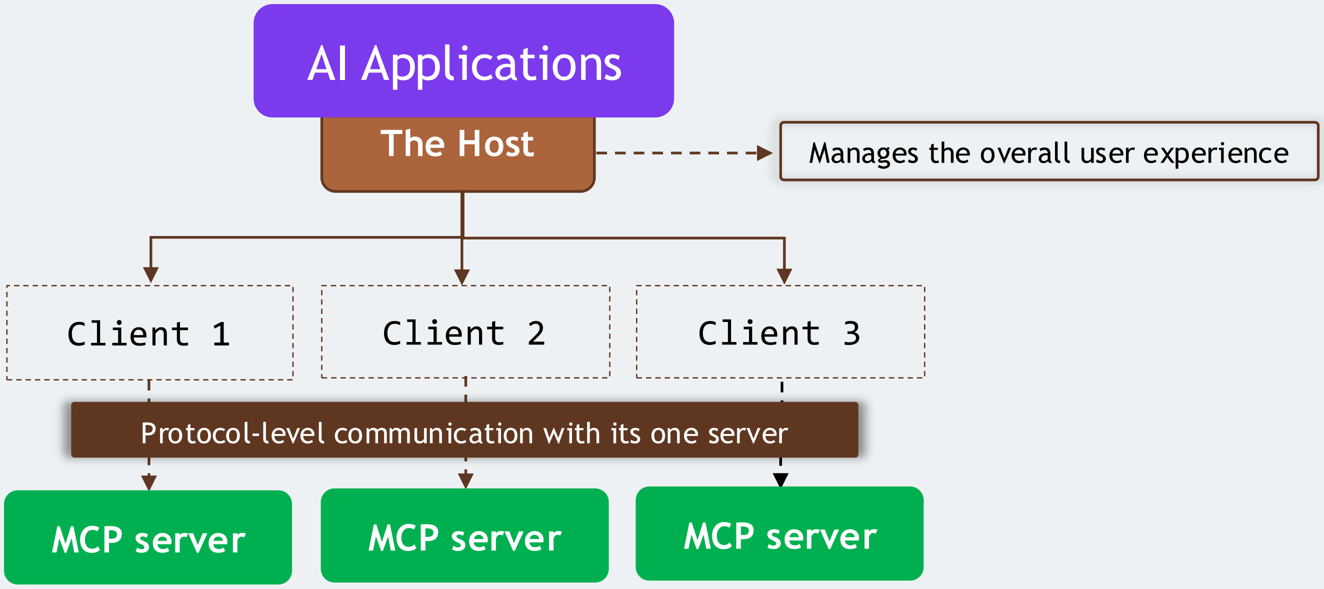
Component that speaks the MCP protocol to a specific server

MCP Client



AI Applications
The Host
Client 1
Client 2
Client 3
MCP server
MCP server
MCP server

MCP Client


AI Applications
The Host
Manages the overall user experience
Client 1
Client 2
Client 3
Protocol-level communication with its one server
MCP server
MCP server
MCP server

MCP Client

Elicitation

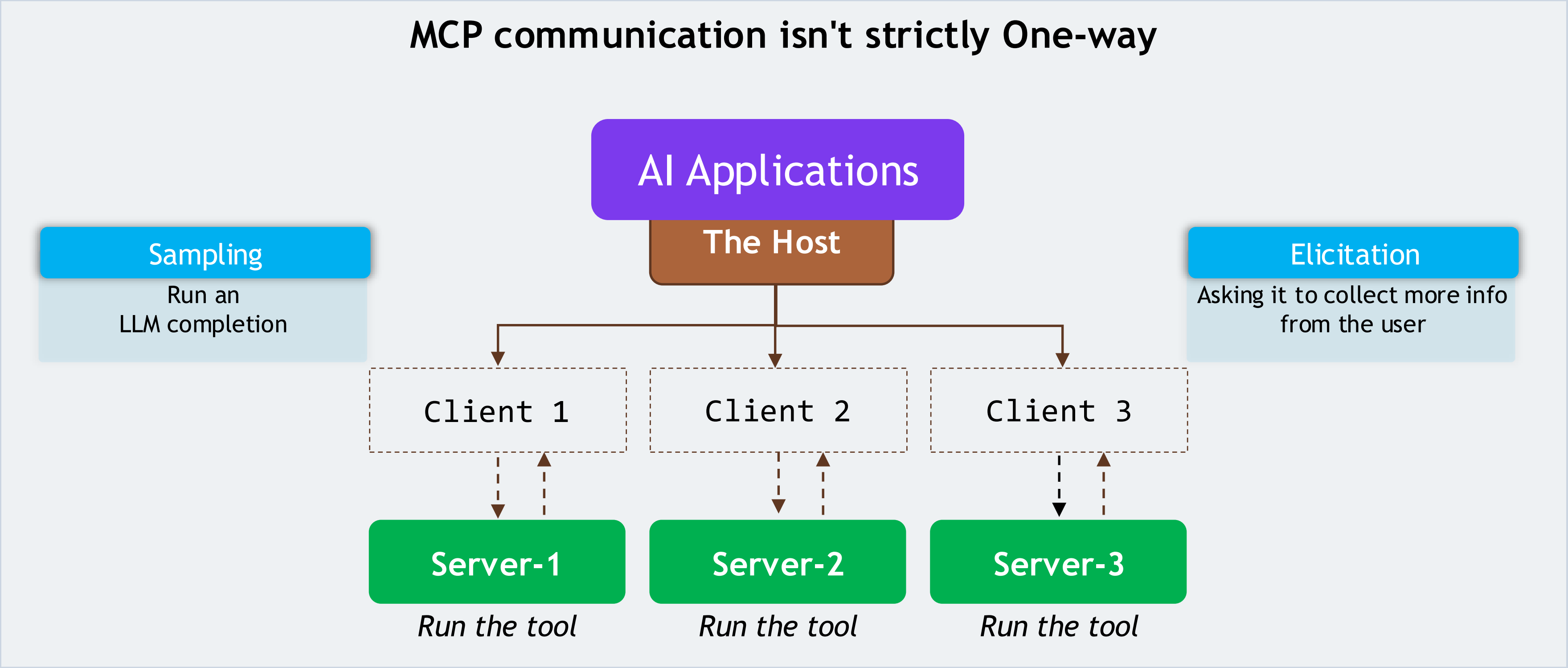
Reusability

MCP Client

Elicitation

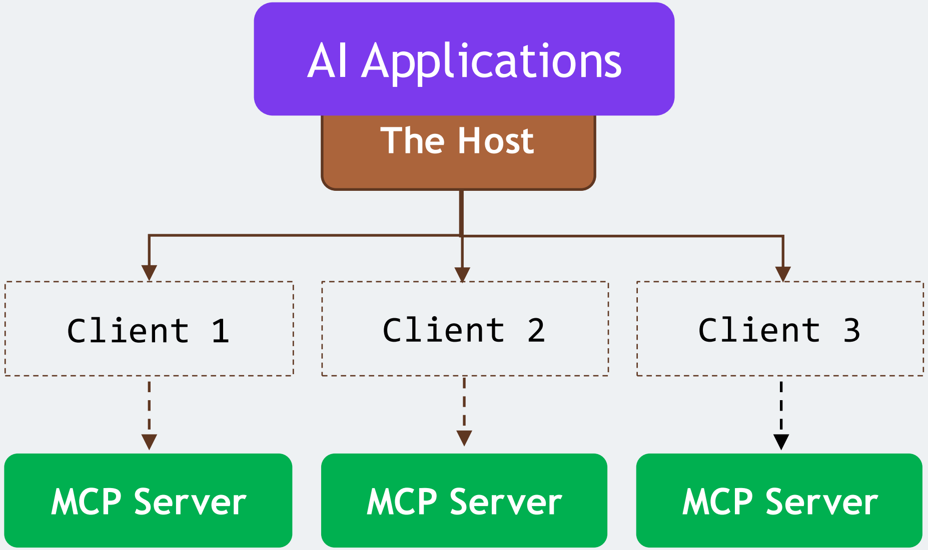
Sampling

MCP Client



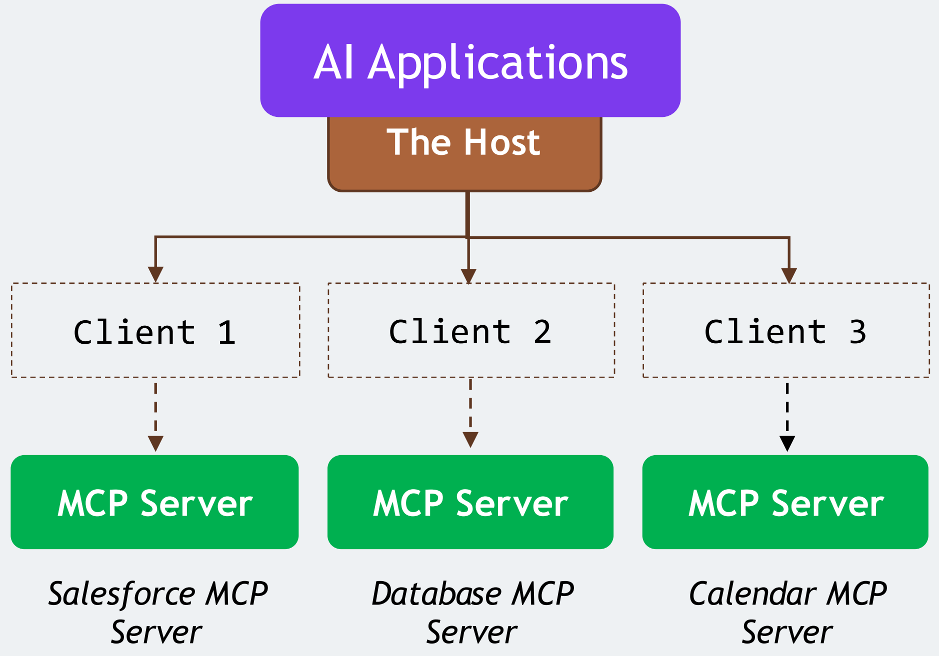
MCP communication isn't strictly One-way
AI Applications
The Host
Sampling
Run an
LLM completion
Elicitation
Asking it to collect more info
from the user
Client 1
Client 2
Client 3
Server-1
Server-2
Server-3
Run the tool
Run the tool
Run the tool

MCP Server

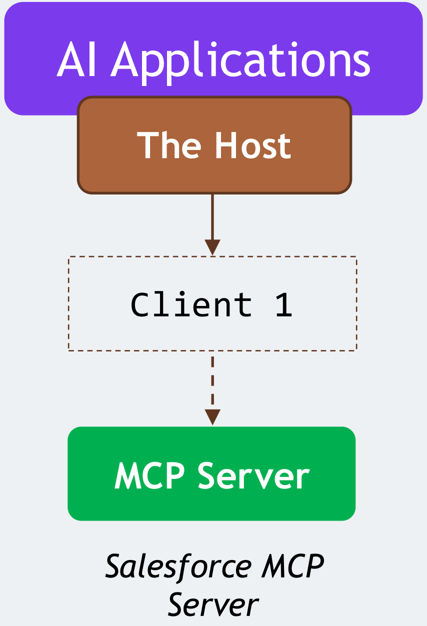


Server is a program that exposes specific capabilities to AI applications through the standardized MCP interface

MCP Server

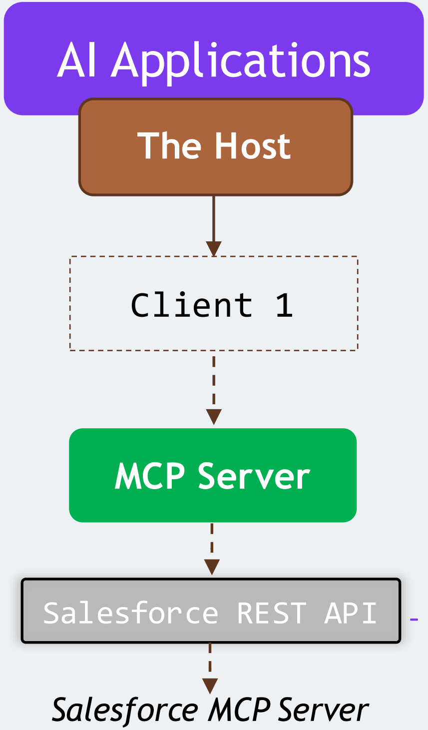

AI Applications
The Host
Client 1
Client 2
Client 3
MCP Server
MCP Server
MCP Server
Salesforce MCP Server
Database MCP Server
Calendar MCP Server

MCP Server



- Server itself isn't necessarily the external system
- Thin standardized layer sitting in front of one

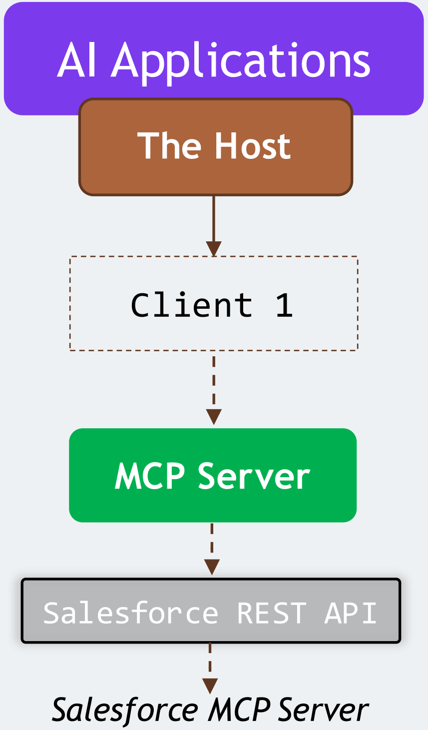
MCP Server



- Server itself isn't necessarily the external system
- Thin standardized layer sitting in front of one

- Doesn't reimplement Salesforce

MCP Server



- Server itself isn't necessarily the external system
- Thin standardized layer sitting in front of one
- Server is the standardized interface the AI application interacts with
- What happens behind it is entirely up to the server's own implementation

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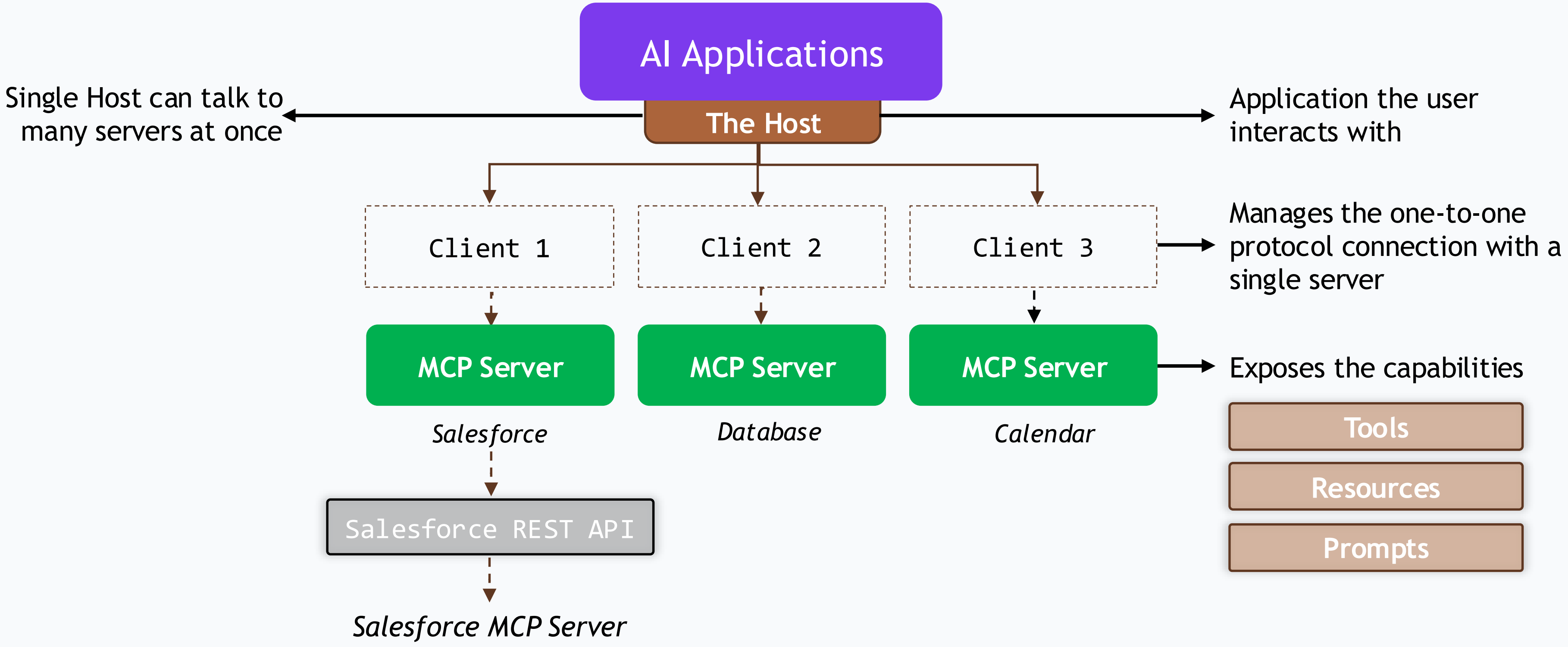
Bringing the Architecture Together

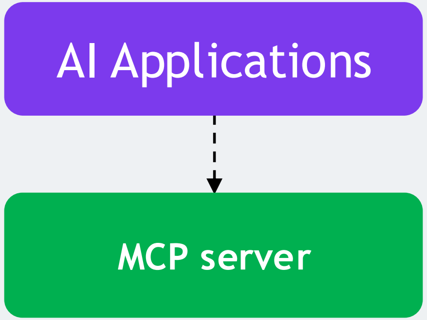


MCP Primitives: Tools, Resources and Prompts

Section 1 › MCP Introduction

MCP Primitives

What exactly can this server provide or expose to an AI application?



MCP Primitives

MCP Primitives

Three core building blocks

- Tools
- Resources
- Prompts

Tools


Tool
AI Applications
MCP server
searchFlights()
checkWeather()
createCalendarEvent()
sendEmail()
searchDatabase()
AI: "What's the weather in Delhi?"
|
v
checkWeather()
|
v
Weather Result
|
v
AI

Tools

Tools

- Tools let the AI do something
- Can call external APIs
- Modify files
- Write to databases
- Trigger workflows
- Perform other operations

Structured description of its parameters

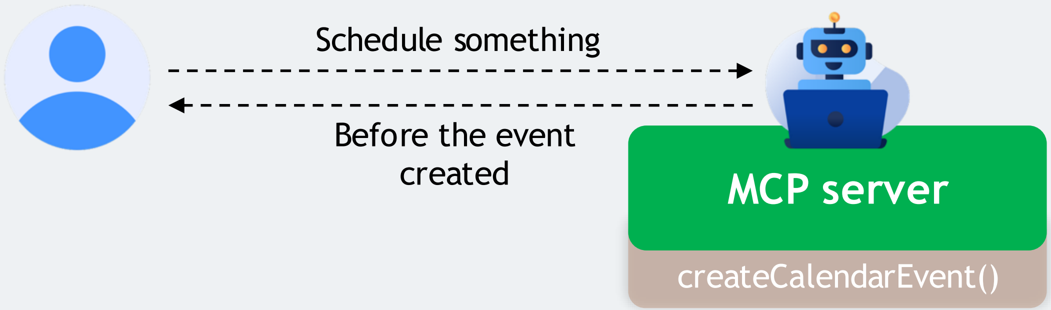
AI model knows exactly what information to provide when it calls a tool like **checkWeather()** or **createCalendarEvent()**

Tools

Applications

Specification describes Tools as model-controlled the AI model can discover and invoke them based on the user's request

Tools


Schedule something
Before the event created
MCP server
createCalendarEvent()

Resources

Resources

AI Applications

Resource provides information or context that an AI application can retrieve

MCP server

Exposes a resource containing a project document

- Project documentation
- Database schema
- Configuration files
- Calendar data

Resources

Resources : Let the AI application access information

```
Tool
 |
 v
Perform an action
```

Model-controlled

```
Resource
 |
 v
Provide information
```

Application-controlled

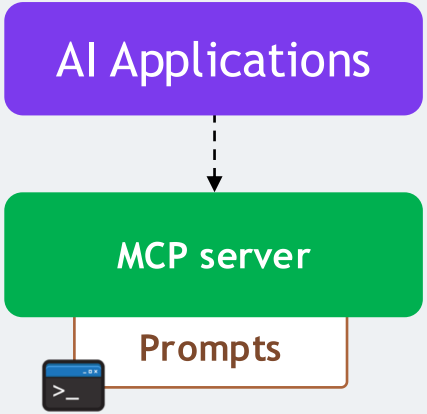
Resources

URI's → Uniform Resource Identifiers

```
file:///Documents/project-plan.pdf
calendar://events/2026
```

- Gives application a standardized way to point to a specific resource
- Each resource also carries a MIME type
- Application knows whether it's plain text, JSON, a PDF, or something else

Prompts




```
/analyze-project
/plan-vacation
/review-code
```

Reusable instruction templates provided by an MCP Server

Important risks | Dependencies | next steps

Prompts

Prompts: Provide reusable instructions or workflows

Prompts are user-controlled they're explicitly selected by the user, not triggered automatically by the model

Prompts

Define its own arguments specific inputs it expects from the user

 ·

Prompts

MCP server

- *Destination: Barcelona*
- *Duration: 7 days*
- *Budget: $3000*

The Actual Difference

```
TOOLS specify         -> "What can the AI DO?"
RESOURCES specify     -> "What INFORMATION can the AI access?"
PROMPTS specify       -> "What INSTRUCTIONS or WORKFLOW can the user invoke?"
```

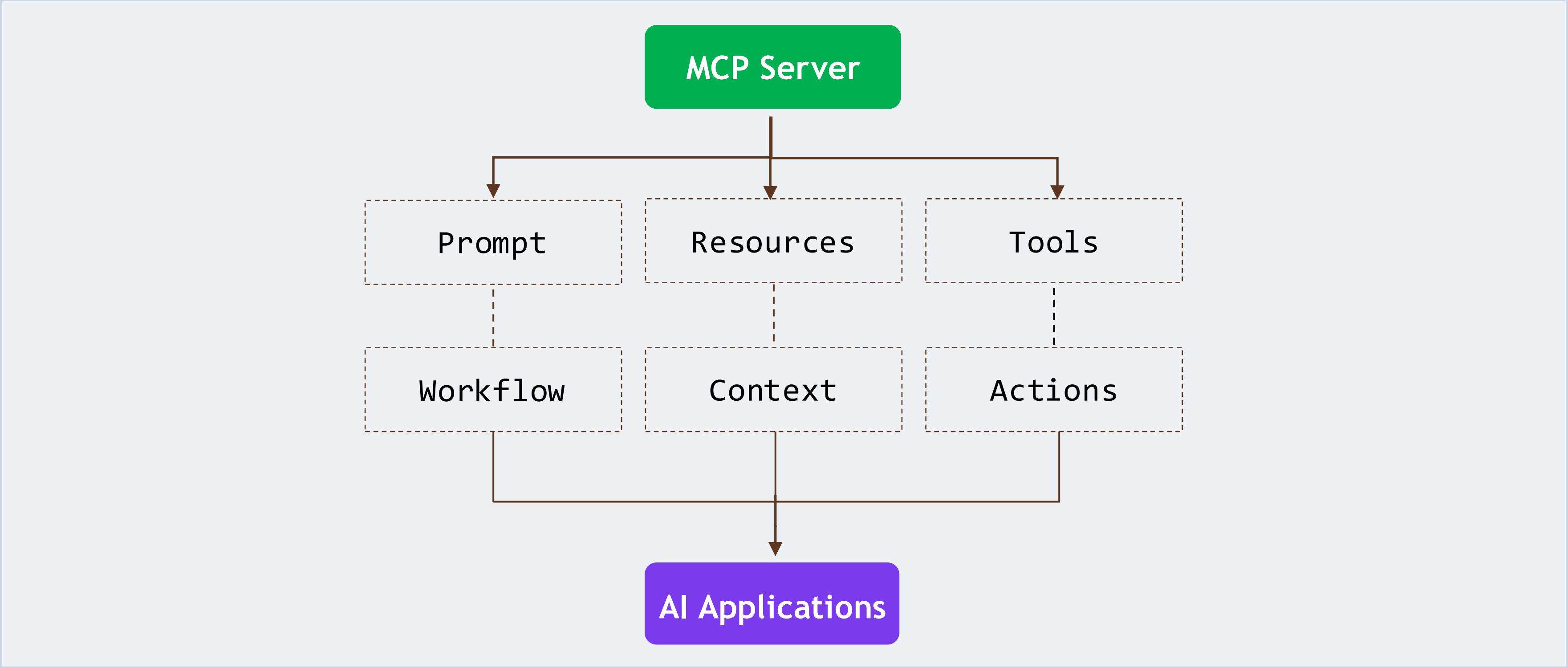
Model-controlled Tools | Application-controlled Resources | User-controlled Prompts

Can They Work Together?

Designed to work together

```
Prompt:      /plan-vacation
Resources:   calendar://my-calendar
             travel://past-trips
Tools:       searchFlights()
             checkWeather()
             bookHotel()
             createCalendarEvent()
```

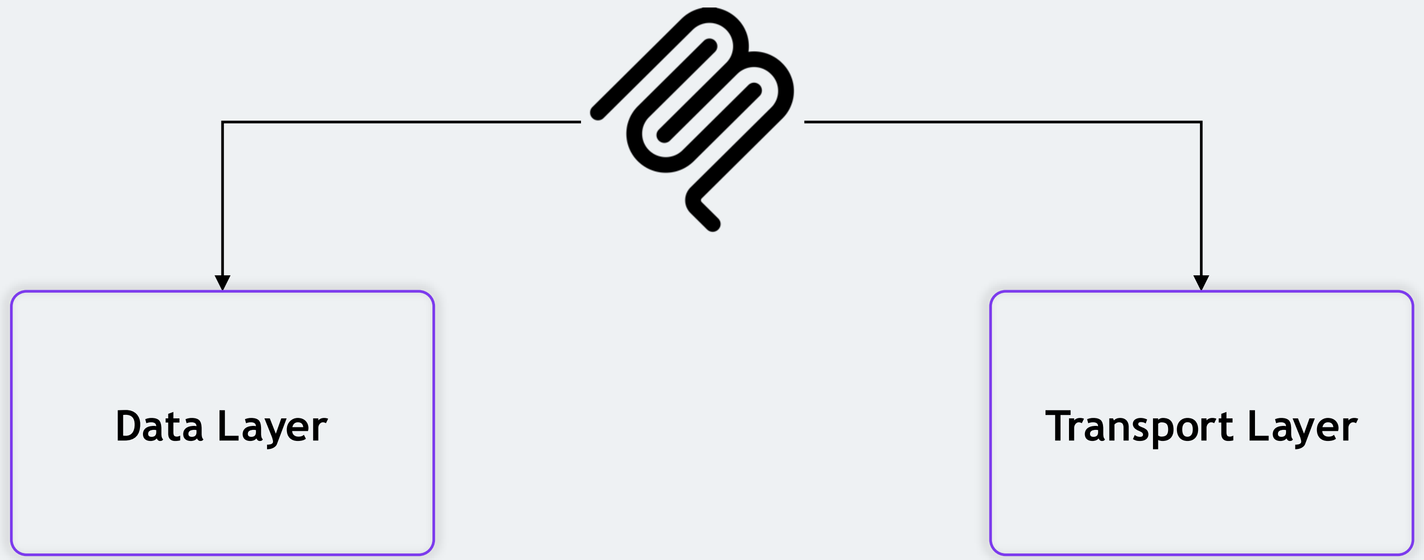
Can They Work Together?



MCP Communication
Data Layer & Transports

Section 1 › Foundations of AI Agents

MCP Communication



Data Layer

Data Layer defines what the Client and Server communicate about, and what those messages mean

```
Client  --(request)-->  Server
```

```
Client  <--(response)--  Server
```

Data Layer

Simple request/response structure that's been a web standard for years

JSON-RPC 2.0 - Tools/list

```
Client  --"Give me the list of available tools"-->  Server
```

```
Client  <--"Here are the available tools"--  Server
```

Data Layer

Data Layer

- Covers requests and responses
- Notifications
- Capability negotiation between Client and Server
- Operations on Tools, Resources, and Prompts

Transport Layer

How does that message physically get from Client to Server?

Transport Layer

- Handles connection setup
- Message framing
- Authorization

Analogy

Writing Letter

Data Layer

- Choosing how to deliver it
- Mail
- Courier
- Hand it over yourself

Transport Layer

Two Standard Transports

stdio

Streamable HTTP

`stdio -> local MCP servers`

- Runs the MCP Server locally
- Client communicates through standard input/output
- No network connection required

`Streamable HTTP -> remote MCP servers`

- Connects to a remote MCP Server
- Uses HTTP for communication and streaming
- Supports standard authentication like OAuth

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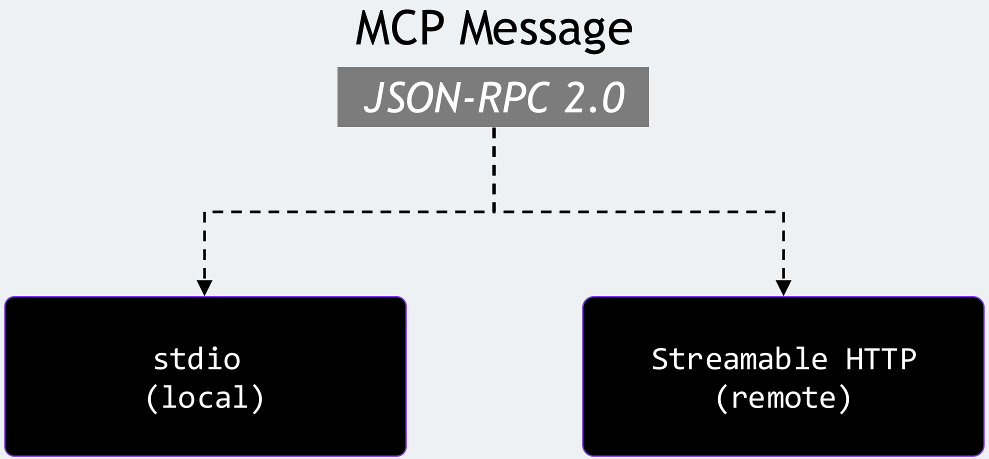
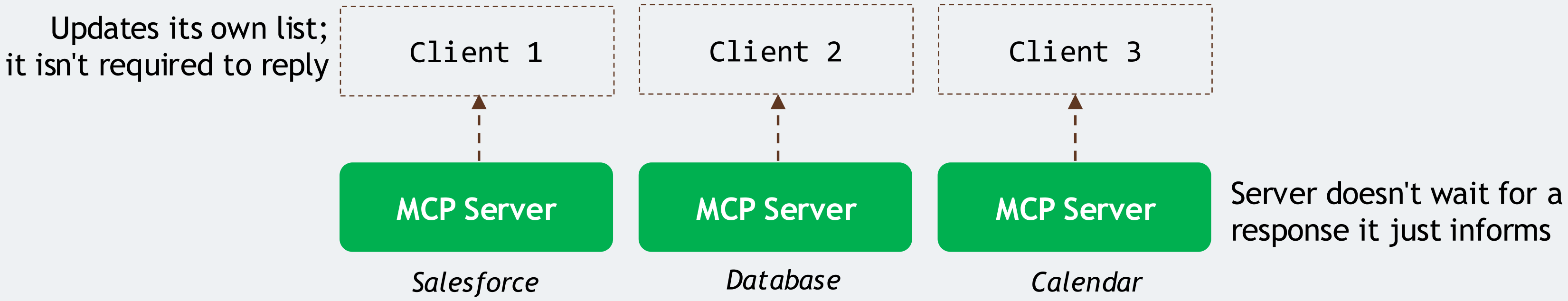
The Important Part



Notifications

Notification is one-way -the sender fires it and moves on, without expecting anything back from the receiver

Notifications



The Important Part

Data Layer

What are we communicating?

How are we communicating?

Transport Layer

MCP Versions, Capabilities & Protocol Evolution

Section 1 › Foundations of AI Agents

Analogy

How does your Client know which version of MCP a given Server supports?

What that Server can actually do?

MCP Client

MCP Versions, Capabilities & Protocol Evolution

Versioning

Capabilities

Protocol Evolution

MCP Versioning

MCP is evolving fast, and version-awareness matters

MCP doesn't use version numbers like MCP 1.0 or MCP 1.1

2026-07-28

Which MCP specification an implementation follows

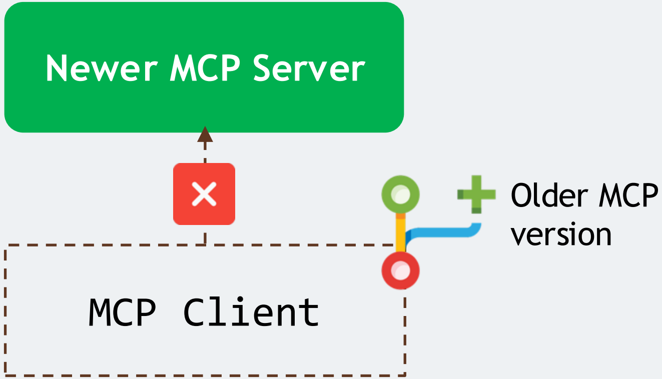
```
MCP Protocol Version   -> 2026-07-28
```

2025-11-25

```
GitHub MCP Server      -> Version 1.4.0
```

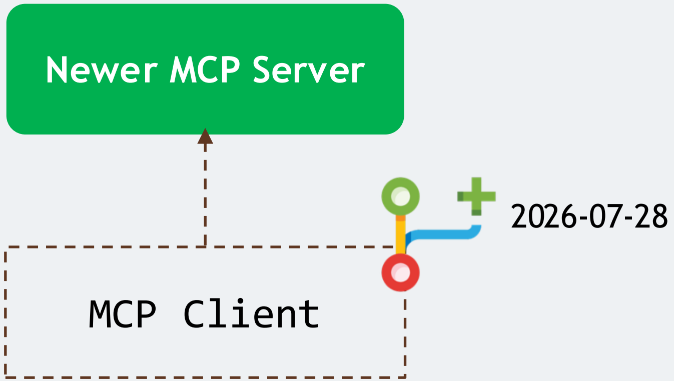
Version of one particular piece of software

Why Do We Need Versioning?



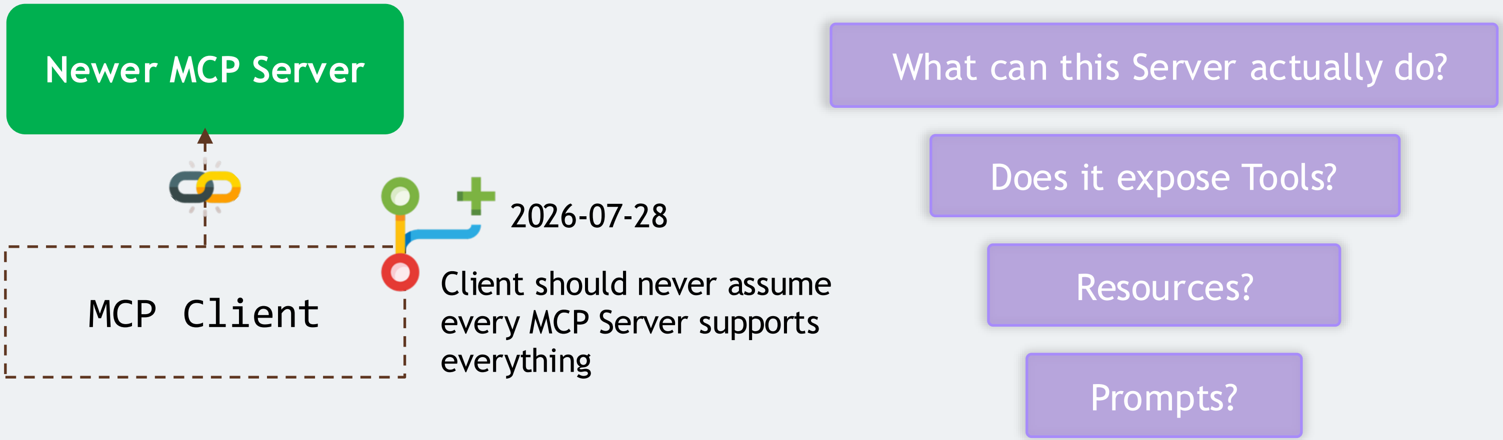
Protocol needs a way to evolve while keeping older and newer implementations interoperable

Why Do We Need Versioning?

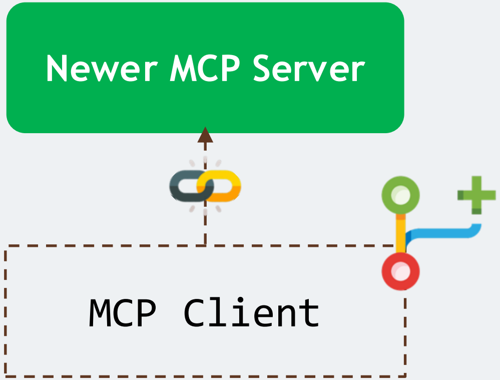


- Formal deprecation policy
- Giving implementations a predictable, guaranteed window
- Migrate whenever a feature is phased out

MCP Capabilities and Discovery



MCP Capabilities and Discovery



2026-07-28

Provides a dedicated RPC for this: **server/discover**

Client can call it when it wants a Server's capabilities up front

MCP Capabilities and Discovery

Newer MCP Server

MCP Client

Stateless core

2026-07-28

- Own protocol version
- Client identity
- Capabilities

a _meta field

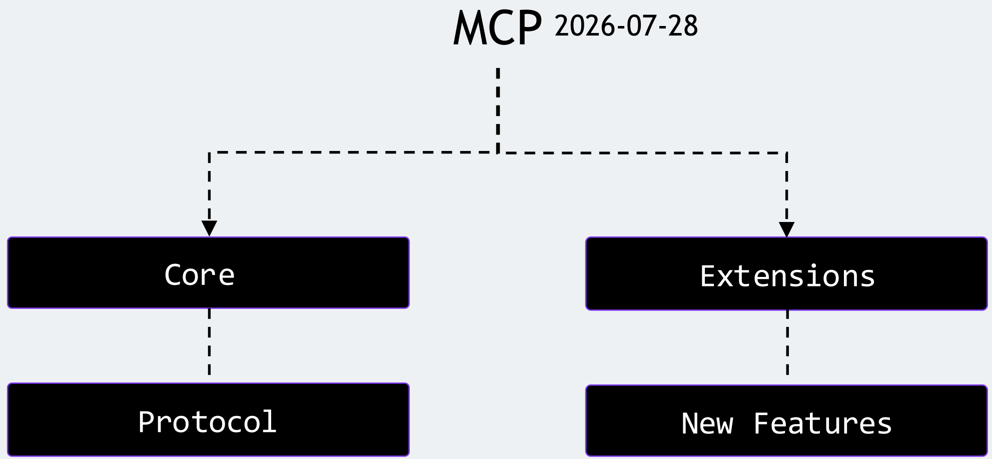
Call server/discover when it wants that information, without ever needing the old initialization step first

MCP Capabilities and Discovery

In Short

Capabilities describe what an implementation supports; discovery is how a Client learns that, on demand

MCP Extensions



What About Deprecated Features?

This stops working tomorrow

This is no longer recommended for new implementations

What About Deprecated Features?

2026-07-28v

Formalizes this with a
guaranteed minimum twelve-month window
between deprecation and removal

What About Deprecated Features?

2026-07-28v

- ✓ **Sampling & Logging** — all still work, but new implementations shouldn't build on them going forward
- ✓ **The legacy HTTP+SSE transport** — deprecated in favor of Streamable HTTP

Important Change

2026-07-28

```
initialize -> initialized -> MCP Session -> Tool Calls
```

Initialize/initialized handshake and the protocol-level Mcp-Session-Id are both gone

Protocol now has a stateless core

Server/discover is available whenever a Client wants a Server's capabilities up front

Practical takeaway

Practical takeaway

Always check which MCP specification version a tutorial, SDK, or server was written against before you rely on it

LIVE DEMO

MCP Documentation

LIVE DEMO

Introduction to n8n
(Low Code - No Code Platform)

LIVE DEMO

Environment for n8n Setup

LIVE DEMO

Workflow Basics

LIVE DEMO

MCP with n8n Workflows

LIVE DEMO

External MCP Server

LIVE DEMO

n8n & Claude
Building & Working Together via MCP

ThinkRook
By Thinknyx Technologies LLP

MCP Bootcamp: Model Context Protocol with n8n & Claude

Yogesh Raheja · Thinknyx Technologies LLP

MCP N8N course

Conclusion

Understand the MCP Protocol

- Why MCP exists
- Host, Client, and Server
- Tools, Resources, and Prompts
- JSON-RPC and transports
- Versioning & deprecation

Build a Practical MCP Toolkit

- Set up n8n self-hosted
- Configure MCP in n8n
- Build your own MCP servers
- Expose real tools
- Connect external MCP servers

Close the Loop with Claude

- Connect Claude to n8n
- Call workflows as tools
- Build workflows with Claude
- Troubleshoot workflows with Claude
- Work directly inside n8n


WHAT'S NEXT?
Don't just learn.
Build something
that matters.
TAKE ACTION
Pick a Tool
or System
Expose
Over MCP
Connect
to Agent
Build &
Automate
aws
THE IMPACT
Build real solutions.
Automate what matters.
Learn by doing.
Create value.

MCP Bootcamp:
Model Context Protocol with n8n & Claude

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MCP n8n Course